

Flutter — Login & Signup with Firebase (Quiz Submission)

This PDF contains:

- 1) Step-by-step setup for Firebase + Flutter (email/password auth).
- 2) Complete example Flutter code (main.dart, login & signup pages, auth service).
- 3) Placeholder output screenshots (login & signup screens).
- 4) Instructions to create a Git repo and push the project, plus a sample GitHub link.

Use this as a template for your quiz. Replace placeholders (projectId, google-services files) with your Firebase project values and run on an emulator or device to capture real screenshots.

Firestore Setup & Project Configuration

Firestore setup notes: 1. In Firestore console create a new project. 2. Add Android app: provide package name (e.g. com.example.flutter_auth_demo). - Download google-services.json and place android/app/. 3. Add iOS app if needed and download GoogleService-Info.plist into ios/Runner. 4. In project-level build.gradle and app-level gradle files, add Firestore plugins per docs. 5. Enable Email/Password Sign-in provider in Firestore Console > Authentication > Sign-in method.

```
# pubspec.yaml (add these dependencies) dependencies:  
  flutter:
```

```
pubspec.yaml (dependencies)  
  firebase_core: ^2.0.0  
  firebase_auth: ^4.0.0
```

- main.dart :

```
import 'package:firebase_core/firebase_core.dart'; import
'package:flutter/material.dart';
import 'login_page.dart';
import 'signup_page.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) { return
    MaterialApp(
      title: 'Flutter Firebase Auth Demo',
      theme: ThemeData(primarySwatch: Colors.blue), home:
      LoginPage(),
      routes: {
        '/signup': (_) => SignupPage(),
      },
    );
  }
}
```

- auth_service.dart :

```
import 'package:firebase_auth/firebase_auth.dart'; class

AuthService {
  final FirebaseAuth _auth = FirebaseAuth.instance;

  Future<UserCredential> signUp(String email, String password) async { return
    await _auth.createUserWithEmailAndPassword(
      email: email, password:
      password,
    );
  }
}
```

```

Future<UserCredential> signIn(String email, String password) async { return
  await _auth.signInWithEmailAndPassword(
    email: email, password:
    password,
  );
}

Future<void> signOut() async {
  await _auth.signOut();
}
}

```

LOGIN PAGE :

```

import 'package:flutter/material.dart'; import
'auth_service.dart';

class LoginPage extends StatefulWidget {
  @override
  _LoginPageState createState() => _LoginPageState();
}

class _LoginPageState extends State<LoginPage> { final
  _emailCtl = TextEditingController();
  final _passCtl = TextEditingController();
  final AuthService _auth = AuthService();
  bool loading = false;

  void _login() async { setState(()
    => loading = true); try {
    await _auth.signIn(_emailCtl.text.trim(), _passCtl.text.trim());
    ScaffoldMessenger.of(context)
      .showSnackBar(SnackBar(content: Text('Logged in')));
  } on Exception catch (e) {
    ScaffoldMessenger.of(context)
      .showSnackBar(SnackBar(content: Text('Error: $e')));
  } finally {
    setState(() => loading = false);
  }
}

```

```

}

@override
Widget build(BuildContext context) { return
  Scaffold(
    appBar: AppBar(title: const Text('Login')),
    body: Padding(
      padding: EdgeInsets.all(16),
      child: Column(
        children: [
          TextField(
            controller: _emailCtl,
            decoration: InputDecoration(labelText: 'Email'),
          ),
          TextField(
            controller: _passCtl,
            decoration: InputDecoration(labelText: 'Password'), obscureText:
            true,
          ),
          SizedBox(height: 16),
          ElevatedButton(
            onPressed: loading ? null : _login, child:
            loading
              ? CircularProgressIndicator(color: Colors.white)
              : Text('Login'),
          ),
          TextButton(
            onPressed: () => Navigator.pushNamed(context, '/signup'), child:
            Text('Create account'),
          ),
        ],
      ),
    ),
  );
}

```

SIGN UP :

```
import 'package:flutter/material.dart';
import 'auth_service.dart';

class SignupPage extends StatefulWidget {
  @override
  _SignupPageState createState() => _SignupPageState();
}

class _SignupPageState extends State<SignupPage> { final
  _emailCtl = TextEditingController();
  final _passCtl = TextEditingController(); final
  AuthService _auth = AuthService(); bool
  loading = false;

  void _signup() async { setState(()
    => loading = true); try {
    await _auth.signUp(_emailCtl.text.trim(), _passCtl.text.trim());
    ScaffoldMessenger.of(context)
      .showSnackBar(SnackBar(content: Text('Account created')));
    Navigator.pop(context);
  } on Exception catch (e) {
    ScaffoldMessenger.of(context)
      .showSnackBar(SnackBar(content: Text('Error: $e')));
  } finally {
    setState(() => loading = false);
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Sign up')),
    body: Padding(
      padding: EdgeInsets.all(16), child:
        Column(
          children: [ TextField(
            controller: _emailCtl,
            decoration: InputDecoration(labelText: 'Email'),
          ),
            TextField(
```

Login

Email

Password

LOGIN

firebase auth demo

Sign up

Email

Password

SIGN UP

firebase auth demo

Sign up

SIGN UP

firebase auth demo

